ZKSF

Zero Kelvin Simulation Foundry

ZHF-v0.1

2026-10-28T11:34:57+00:00

Certificate of Hardware Fidelity

Certificate ID

e2b1fe6a617649ec

Measured hardware fidelity: 0.4367 vs. exact ideal

ENGINE
qpu.iqm.garnet

METHOD
quantum hardware (IQM Garnet (superconducting))

SHOTS
560

DEVICE
Garnet

HELLINGER FIDELITY
0.4367

VERIFICATION MODE
direct

MEASURED OUTCOMES (top)

000	148 shots
111	99 shots
101	68 shots
001	61 shots
011	58 shots
010	54 shots

HOW THIS CERTIFICATE WAS PRODUCED

• This certificate reports ZHF-v0.1, the hardware fidelity protocol: how closely a real quantum processor's measured output matches the ideal distribution for the same circuit, as opposed to the accuracy of a classical approximation.

• Verification mode is direct: the same circuit was additionally simulated exactly on a classical computer (feasible here because it uses at most 24 qubits). The reported Hellinger fidelity is the statistical overlap between the hardware's measured counts and that exact ideal distribution, on a scale from 0 (no overlap) to 1 (identical).

• Counts are raw hardware output: no error mitigation, post-selection, or readout correction has been applied. Any deviation from the ideal distribution reflects the physical noise of the device.

Independently verifiable at:

https://api.zksf.org/certify/e2b1fe6a617649ec

Circuit SHA-256 99b41c00d8345c3a2075bce453e75bcb43a1a6af0dc51855bf759b4482643fee